WINE QUALITY PREDICTION

Using Decision Trees and Random Forest



Our project uses Machine Learning to predict wine quality based on its chemical properties. It uses two algorithms: Decision Tree and Random Forest.
We use two algorithms, Decision Tree and Random Forest, and compare their performance to find which gives better predictions.

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Introduction

From chemical measurements to an intelligent quality decision

WHY WINE QUALITY MATTERS

Wine quality is traditionally assessed by experts using taste, aroma, colour and texture. Your project uses measurable chemical properties to help predict quality.
The model considers 11 features, such as acidity, sugar, density, pH, sulphates and alcohol.

WHAT THE MODEL USES

- 11 physicochemical features
- Examples: acidity, sugar, chlorides, density, pH, sulphates and alcohol
- Historical wine data is used to learn patterns

WHY MACHINE LEARNING

- Supports data-driven prediction
- Reduces dependence on manual tasting alone
- Helps make quality assessment faster and more consistent

Decision Tree = interpretable rules | Random Forest = ensemble of trees

Problem Statement

Why is automated wine quality prediction needed?

EXISTING PROBLEM

Manual wine evaluation depends on sensory judgement by experts. The same sample may receive different ratings from different people.

LIMITATIONS

- Can be time-consuming
- May increase evaluation cost
- Human judgement can vary
- Difficult to keep assessments consistent

WHO IS AFFECTED?

- Wineries and producers
- Quality-control laboratories
- Businesses that need repeatable quality checks

Need: a reliable, data-driven system that predicts quality from chemical measurements.

Motivation

Why this project is useful in a real-world setting

CONSISTENCY

Our main motivation is to improve consistency, save time and compare two different Machine Learning algorithms.
By using chemical data, the system can provide a useful quality estimate to support decision-making.

EFFICIENCY

Automated predictions can save time and support quality-control workflows.

MODEL COMPARISON

Compare a simple, interpretable Decision Tree with a Random Forest ensemble.

Project motivation: turn raw chemical measurements into useful quality predictions.

Objectives

Main goal and specific objectives

MAIN OBJECTIVE

Build a reliable machine-learning system to classify wine quality using physicochemical measurements.

- Prepare the dataset by handling missing values and duplicate records.
- Train and compare Decision Tree and Random Forest classifiers.
- Tune model hyperparameters using GridSearchCV.
- Evaluate models using standard classification metrics and visualisations.
- Provide an interactive Streamlit interface for quality prediction.

Existing System

Current approach and its limitations

TRADITIONAL APPROACH

- Experts assess taste, aroma, colour and texture.
- Decisions depend on experience and sensory evaluation.
- Human evaluation remains valuable, but may vary from person to person.

CHALLENGES

- Time and cost of repeated assessment
- Subjective differences in ratings
- Harder to standardise across many samples

The project uses chemical data to support, not replace all expert judgement.

Existing approach: sensory assessment → Proposed support: data-driven prediction

Proposed System

A machine-learning pipeline with an interactive prediction interface

INPUT

User enters 11 wine physicochemical properties.

PREDICTION

Trained Decision Tree and Random Forest models process the input.

OUTPUT

Streamlit displays a predicted class: Poor, Average or Good.

How it addresses limitations: standardised inputs, repeatable model predictions, and measurable evaluation.

System Architecture / Block Diagram

End-to-end flow from dataset to user prediction



INPUT: 11 physicochemical properties → MODEL: Decision Tree + Random Forest → OUTPUT: Poor / Average / Good

Methodology / Workflow

Step-by-step process followed in the project

1 **Load**

Read the wine dataset.

2 **Prepare**

Clean the data and select features.

3 **Split**

Divide the data into training and testing sets.

4 **Train**

Teach both models using training data.

5 **Tune**

Find suitable model settings.

6 **Evaluate**

Measure performance using metrics and graphs.

7 **Deploy UI**

Allow users to make predictions through the interface.

Train/test split and cross-validation help estimate how well the model generalises.

Technology Stack

Software and libraries listed in the project abstract

Pandas — The programming language used for the Machine Learning work.

NumPy — Performs numerical calculations.

Scikit-learn — Loads, cleans and organises tabular data.

Matplotlib — Create graphs and visualisations.

Seaborn — Create graphs and visualisations.

Streamlit — Creates the interactive interface for the project.

Joblib — Saves and loads trained models.

Programming language: Python | UI framework: Streamlit | Database: not specified in the supplied abstract

System Modules / Components

Functional parts of the proposed system

Data input

Collect 11 chemical measurements.

Preprocessing

Clean records and remove duplicates.

Model training

Train Decision Tree and Random Forest.

Tuning & validation

GridSearchCV and cross-validation.

Evaluation

Metrics, confusion matrix, ROC and AUC.

Prediction UI

Streamlit accepts inputs and shows class.

Dataset / Input Data

Wine Quality dataset snapshot

1,599	240	1,359	11
Original samples	Duplicate records removed	Remaining samples	Input features

11 INPUT FEATURES

Fixed acidity • Volatile acidity • Citric acid • Residual sugar • Chlorides • Free SO_2 • Total SO_2 • Density • pH • Sulphates • Alcohol

TARGET CLASSES

POOR: score ≤ 4
AVERAGE: score 5–6
GOOD: score ≥ 7

Target variable: wine quality score

Algorithms / Model

Why these supervised learning algorithms were selected

DECISION TREE

- Learns a tree of decision rules
- Easy to understand and interpret
- A single tree may overfit complex data

RANDOM FOREST

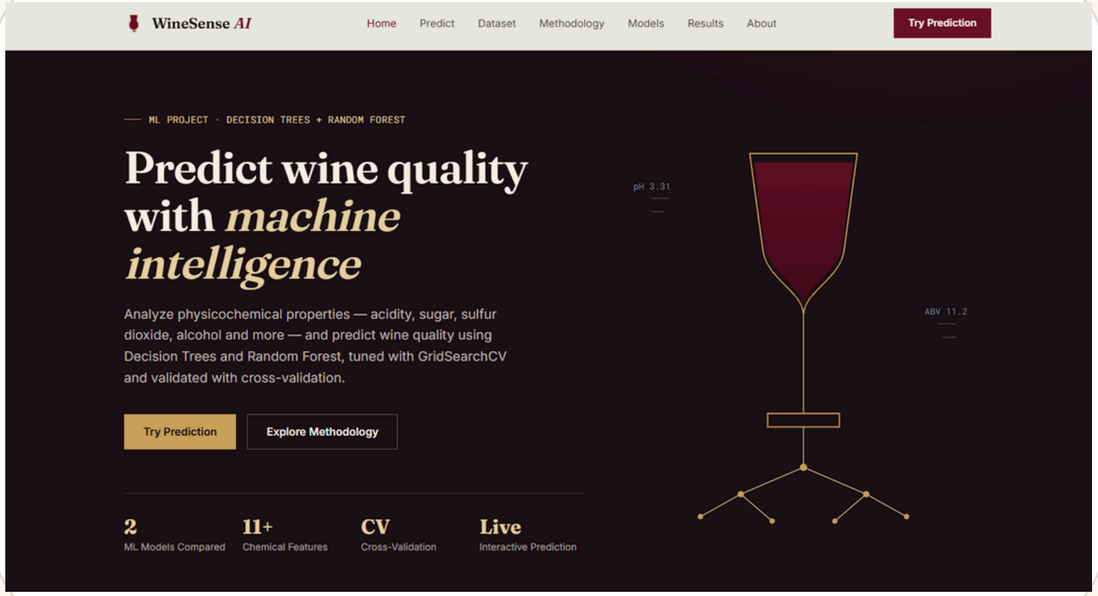
- Combines predictions from multiple decision trees
- Usually provides more stable predictions
- Helps reduce overfitting compared with one tree

Model selection is based on comparing measured performance, not only assumptions.

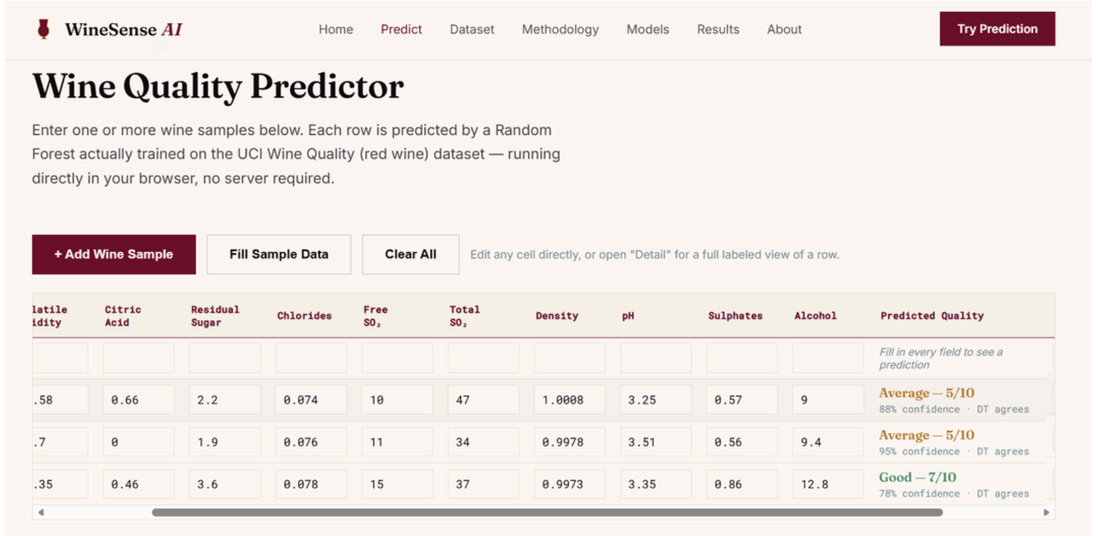
Implementation & Website Demo

Screenshots from the submitted WineSense AI website

01 HOME PAGE



02 PREDICTION DASHBOARD



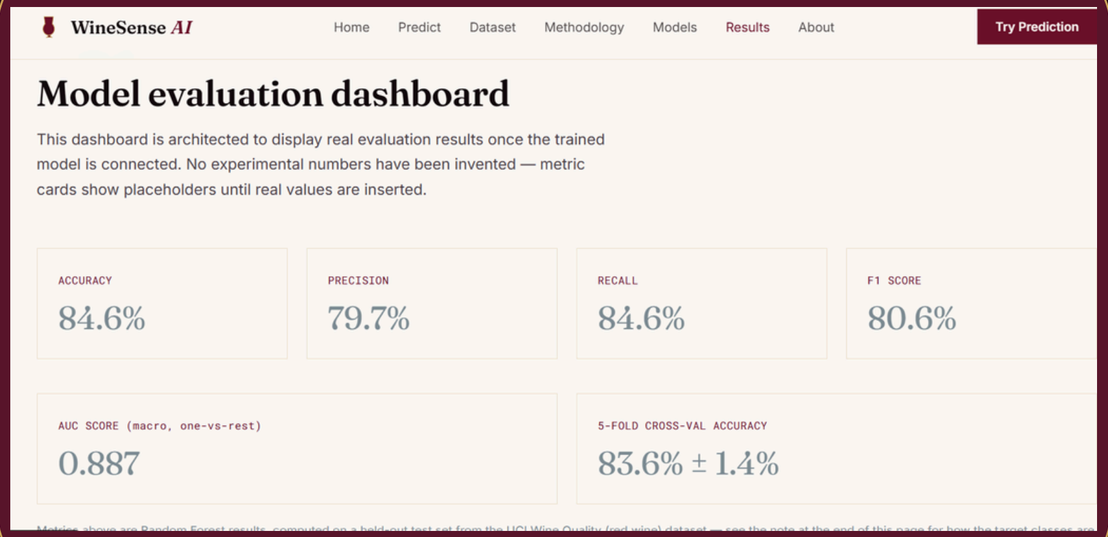
Home: introduces WineSense AI and offers navigation to prediction, methodology, models, results and project information.

Predict: users enter 11 physicochemical properties. The sample rows show the predicted quality class and confidence.

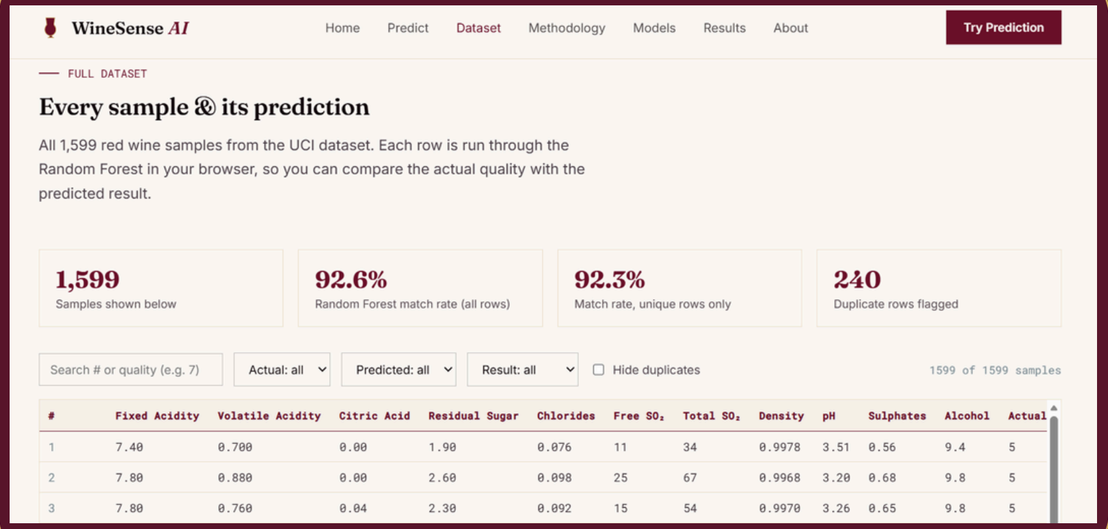
Website Demo: Results & Dataset

Evaluation dashboard and dataset explorer

03 MODEL EVALUATION



04 DATASET EXPLORER



Results: displays accuracy, precision, recall, F1-score, AUC and cross-validation metrics. Verify these values against your final model run before presenting.

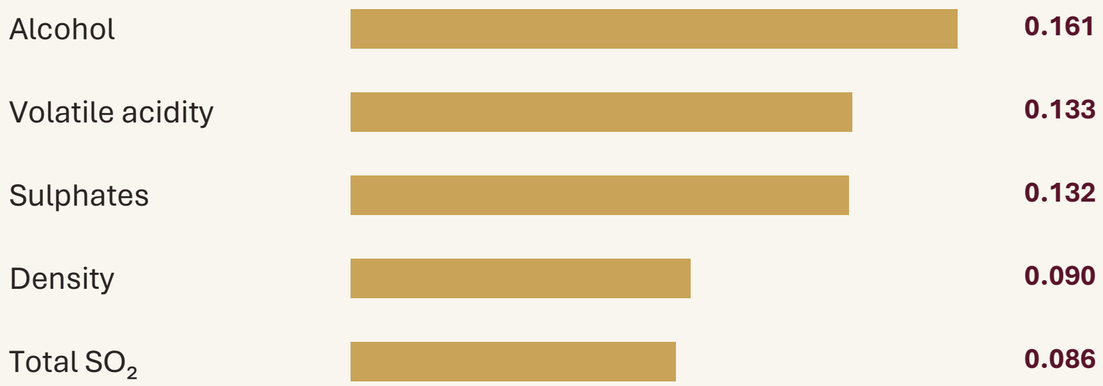
Dataset: shows 1,599 red-wine samples, match-rate summaries, filters and measured chemical features.

Results

Performance values shown in the review presentation

84.6%	79.7%	84.6%	0.887
Accuracy	Precision	Recall	AUC

FEATURE IMPORTANCE



KEY RESULT

The review deck reports that Random Forest performs better overall than the single Decision Tree. The app is described as showing predictions, confusion matrix, ROC curve, feature importance and model comparison.

F1-score was requested in the faculty checklist but no value was provided in the supplied files.

Advantages, Applications & Limitations

Where the system can help—and what to keep in mind

ADVANTAGES

- Faster assessment support
- More consistent predictions
- Comparison of two models
- Evaluation with standard metrics

APPLICATIONS

- Winery quality-control workflows
- Quality-control laboratories
- Data-supported production decisions

LIMITATIONS

- Predictions depend on dataset quality
- Chemical inputs alone may not capture every sensory factor
- The supplied files do not quantify deployment or external-test performance

Future Scope

Possible improvements and extensions

- Evaluate on additional and more diverse wine samples.
- Continue tuning model parameters and compare further classifiers.
- Improve the Streamlit interface and make result visualisations clearer.
- Deploy the application so users can access it more easily.
- Add model monitoring and periodic evaluation as new data becomes available.

These are proposed future directions, not features claimed as already completed.

Conclusion

Turning wine chemistry into a data-driven quality estimate

This project applies supervised machine learning to predict wine quality from 11 physicochemical properties.

WHAT WAS BUILT

A pipeline for data preparation, model training, evaluation and web-based prediction.

WHAT WAS COMPARED

Decision Tree and Random Forest classifiers.

KEY OUTCOME

The review deck reports stronger overall results for Random Forest.

Thank You